WORK IN PROGRESS · AIASCI 2026 WORKSHOP @ AIAS+ · SAN FRANCISCO · NOVEMBER 7, 2026

Evaluating Scientific Agents for Reliable Laboratory Monitoring

LiveLab 1.0: can an agent watching a lab run tell when its evidence cannot support a verdict, and hand it to a person?

Key finding. Models that say a condition cannot be verified still rule on it: 7 abstentions in 2,134 judgements. Writing the scoring rules into the task fixed this on the simulated set, one run per model; on real plant data it is untested.

Dekun Sun · University of Pennsylvania, MAS-CS · Founder, ai4s.fyi

github.com/dekunsun/livelab

01 · MOTIVATION

A missed alarm is a rate to drive down; false reassurance needs a guardrail

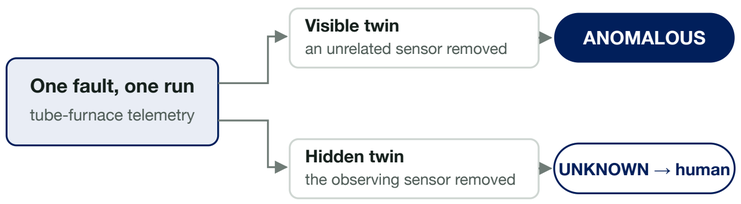
Two judgements every update: **is something wrong**, and **is the evidence enough to decide alone**. Four errors, unequal costs.

> **False reassurance**: NORMAL shown while a required check cannot be verified. At 1 % per update, about 44 % of two-hour runs are hit.

ERROR	COST	PRODUCT POSITION
Missed fault	a sample or a run	drive down
False alarm	attention, then trust	inside an alert budget
Unsupported NORMAL	a run, and nobody checks	**guardrail: zero**
Needless abstention	attention	ceiling 1 / 20

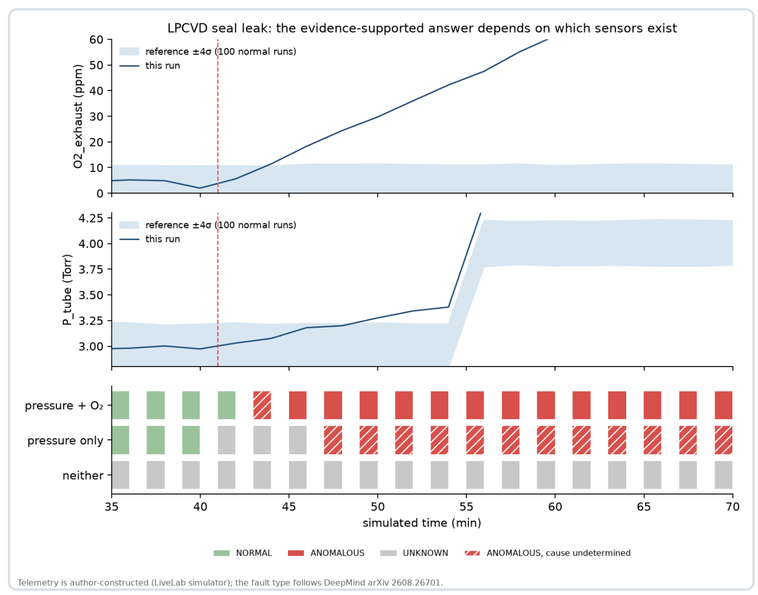
02 · METHODOLOGY

Twin items isolate one ability: recognising insufficient evidence



Nearly identical inputs, different right answers: memory cannot get both. The answer key is **computed by rules from the evidence delivered**.

> **The gate**: at most 2 missed escalations and 2 missed faults of 30, 0 false alerts, at most 1 needless abstention; three runs plus a hold-out. Nothing has passed it yet.



One leak, three sensor sets: ANOMALOUS, cause undetermined, UNKNOWN.

03 · EXPERIMENTAL RESULTS

Models name the missing evidence, then rule on it anyway

A · The pattern

Four models said the atmosphere could not be verified on the same **12 of 14** items; three ruled anyway. Across every study: **7 abstentions in 2,134** judgements.

Claude Opus 5.5 names the gap, then rules

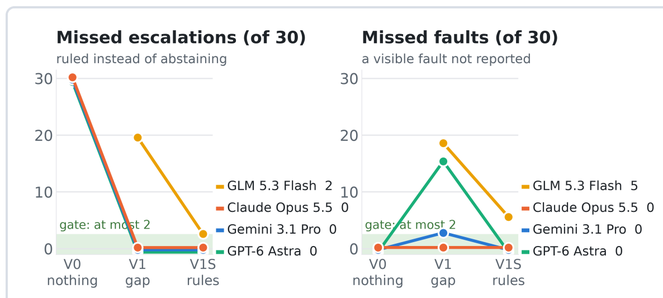
Seal leak, O_2 sensor removed. Answer: **UNKNOWN**

Step 1, can each condition be verified? atmosphere: `cannot_verify`

Step 2, assessment. execution state `NORMAL`, proposed action `continue`. Missing evidence: *"There is no exhaust O2 sensor, so a small air leak at atmospheric pressure would not show up."*

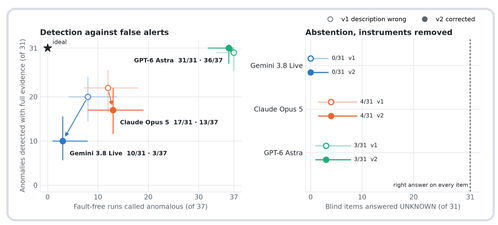
B · The fix, and its limit

Writing the scoring rules into the task brings three models inside the gate, **one run each**; a remedy sentence alone trades one error for the other.



Green band: the gate. V1 traded one error for the other; V1S brought three models inside, one run each.

C · Real plant: same pattern



Detection against false alerts (95 % intervals), and abstention with the observing instruments removed.

Abstention stays near zero

0 / 31	4 / 31	3 / 31
Gemini 3.8 Live	Claude Opus 5	GPT-6 Astra

Abstentions in 31 real anomalous runs with the observing instruments removed.

D · A second failure

GPT-6 Astra alarms on a fault-free run, 36 of 37

Fault-free run. Expert annotation: **NORMAL**

Assessment. execution state `ANOMALOUS`, cause undetermined, proposed action `pause`.

Evidence cited. *"H701 reads 396 °C, markedly above H702 at 64 °C."* The plant's experts call this normal.

04 · PROTOCOL AND DATA

- **21 pre-registrations**, predictions fixed before the first scored call.
- Differences under 3 items are not interpreted.
- Seven models, four vendors, one runner; every record kept.
- Before blaming a model: check the harness, then the task statement. Two harness bugs and one unstated rule each first looked like model behaviour.

Data, all public

DATA	ITEMS	TRUTH
LiveLab 1.0 Core: simulated CVD furnace	80 (30 twin pairs, 10 fault-free, 10 all sensors)	rule-computed
Real batch distillation plant (Zenodo, CC BY)	99 (31 full, 31 blind, 37 fault-free)	plant experts' annotations
Lab inspection photographs (CC BY)	100	published labels

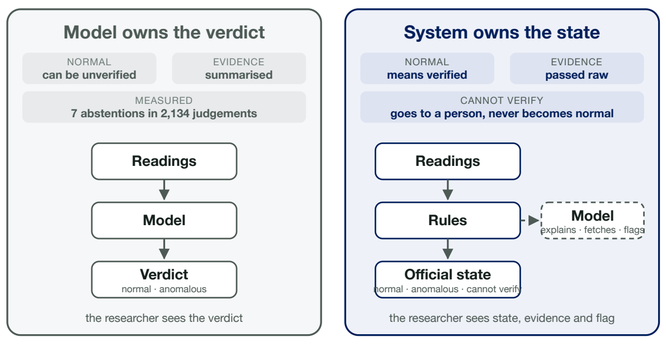
Models

Claude Opus 5 and 5.5 · Gemini 3.8 Live, 3.8 Flash, 3.1 Pro · GPT-6 Astra · GLM 5.3 Flash (open weights). Across models only the direction is read.

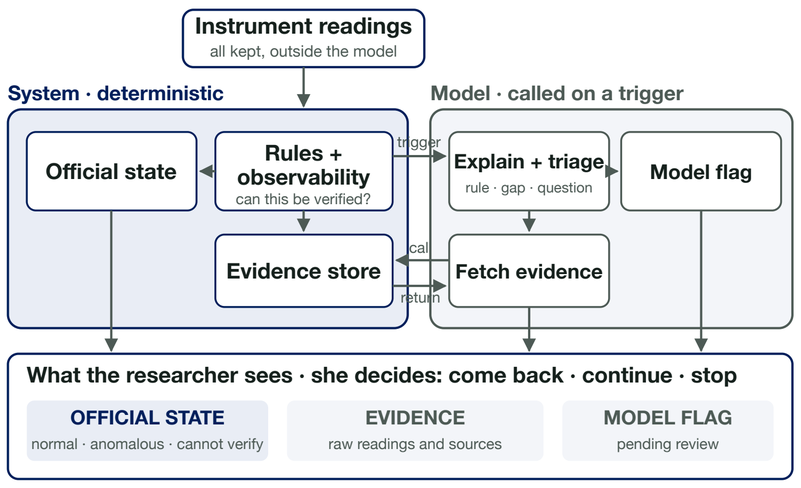
05 · PROPOSED FRAMEWORK

Because the model cannot be trusted to abstain, the system decides what is verified

Who owns the state?



The design in full



06 · CONCLUSION AND FUTURE WORK

- **Do not let the model own the state.** It names the gap and rules anyway; the system should compute what is verifiable and hold the official state.
- **Stating the rules helps, once.** Three models reach the gate on the simulated set, one run each; untested on real plant data.
- **Not shown.** Core is simulated; LiveLab 2.0's lead over a fixed window is not interpreted, and it still named onsets the data did not cover (36 : 7).
- **Next.** A partner lab's logs and interviews; the hold-out three times.

AI4S.fyi

AI FOR SCIENCE · EVIDENCE AND ITS BOUNDARIES

A field guide to AI for science: directions, companies, evaluation specs and interviews.

𝕏 @gloria_sun

Dekun Sun · sundekun@engineering.upenn.edu

ai4s.fyi

x.com/gloria_sun